Function Title: read_fasta

Description:
Reads a FASTA file and stores all sequences in a dictionary with keys as sequence IDs.

Input 1: Path to FASTA file
Output: Dictionary
(Potential) Output: Error message if file cannot be opened

Process:
Step 1: Open the FASTA file.
Step 2: Read line-by-line; detect header lines.
Step 3: Store sequence lines associated with each header.
Step 4: Validate: ensure no empty sequences.
Step 5: Return dictionary of sequences.

Function Title: find_longest_orf

Description:
Detects the longest open reading frame (ORF) within a nucleotide sequence.

Input 1: Nucleotide sequence (string)
Output: Longest ORF subsequence
(Potential) Output: Error message if no valid ORF found

Process:
Step 1: Initialize variable .
Step 2: For each reading frame (0, 1, 2), scan for "ATG" start codons.
Step 3: For every start codon, search forward until a stop codon (TAG/TGA/TAA).
Step 4: Track the longest valid ORF discovered.
Step 5: Return longest ORF or warning if none exists.

Function Title: translate_to_amino_acids

Description:
Translates a nucleotide ORF into an amino acid sequence using a codon table.

Input 1: ORF nucleotide sequence
Output: Amino acid sequence
(Potential) Output: Warning for codons containing ambiguity (N, R, Y, etc.)

Process:
Step 1: Initialize amino acid string.
Step 2: Read codons in steps of 3 bases.
Step 3: Translate each codon using codon translation table.

Step 4: If codon is invalid, assign “X” (unknown).
Step 5: Return amino acid sequence.

Function Title: compute_kmer_similarity_matrix

Description:
Computes pairwise similarity scores between all amino acid sequences based on shared k-mers.

Input 1: Dictionary of amino acid sequences
Input 2: k value (e.g., 3 or 4)
Output: k-mer similarity matrix
(Potential) Output: Error if k is larger than any amino acid sequence

Process:
Step 1: Extract all k-mers from each sequence.
Step 2: For each pair of sequences, count shared k-mers.
Step 3: Normalize similarity score.
Step 4: Fill similarity matrix.
Step 5: Return matrix.

Function Title: sort_sequences_by_similarity

Description:
Sorts sequences into a progressive alignment order using k-mer similarity scores.

Input 1: k-mer similarity matrix
Output: Ordered list of sequence IDs
(Potential) Output: Warning if scores are tied

Process:
Step 1: Select a starting sequence (highest average similarity).
Step 2: Repeatedly add the next most similar unaligned sequence.
Step 3: Build complete ordering list.
Step 4: Return ordered sequence list.

Function Title: needleman_wunsch_alignment

Description:
Performs global alignment between two amino acid sequences using the Needleman–Wunsch algorithm.

Input 1: Amino acid sequence A
Input 2: Amino acid sequence B
Output: Aligned sequence A, aligned sequence B
(Potential) Output: Error if sequences are empty

Process:
Step 1: Initialize scoring and traceback matrices.
Step 2: Fill matrices using match, mismatch, and gap penalties.
Step 3: Trace back from bottom-right corner to top-left.
Step 4: Construct aligned versions of both sequences.
Step 5: Return aligned sequences.

Function Title: progressive_alignment

Description:
Aligns multiple sequences by progressively adding sequences in sorted order.

Input 1: Ordered list of sequence IDs
Input 2: Dictionary of amino acid sequences
Output: Multiple sequence alignment (amino acids)
(Potential) Output: Warning if alignment grows too gappy

Process:
Step 1: Align the first two sequences with Needleman–Wunsch.
Step 2: Treat resulting alignment as a profile.
Step 3: Align each additional sequence to the profile one-by-one.
Step 4: Maintain column consistency.
Step 5: Return final amino acid alignment.

Function Title: back_translate_alignment

Description:
Converts an amino acid alignment back into nucleotide codons while preserving alignment gaps.

Input 1: Aligned amino acid sequence
Input 2: Original nucleotide ORF sequence
Output: Codon-aware aligned nucleotide sequence
(Potential) Output: Error for mismatched ORF length

Process:
Step 1: Initialize pointer at start of ORF.
Step 2: For each amino acid in the aligned sequence:
- If amino acid is "-" → append "---".
- Else append next codon (3 nt) from ORF.

Step 3: Continue until alignment ends.
Step 4: Return codon-preserving aligned sequence.

Function Title: compute_codon_position_statistics

Description:
Computes per-position variability statistics for codon positions 1, 2, and 3 across the aligned nucleotide MSA.

Input 1: Dictionary of aligned nucleotide sequences
Output: Table/CSV of variability at each codon position
(Potential) Output: Warning if alignment length is not divisible by 3

Process:
Step 1: Walk alignment in steps of 3 nucleotides.
Step 2: For each codon column, compute variability at positions 1, 2, and 3.
Step 3: Store variability scores in a table.
Step 4: Return results as a dataframe or dictionary.

Function Title: generate_plots

Description:
Creates visualizations (PNG) of codon variability or other alignment statistics.

Input 1: Codon statistics table
Output: PNG plot stored in output directory
(Potential) Output: Error if plotting library not installed

Process:
Step 1: Import matplotlib or another plotting library.
Step 2: Create a figure showing variability versus codon position.
Step 3: Label axes, title, legend.
Step 4: Save plot to file.
Step 5: Return success message.

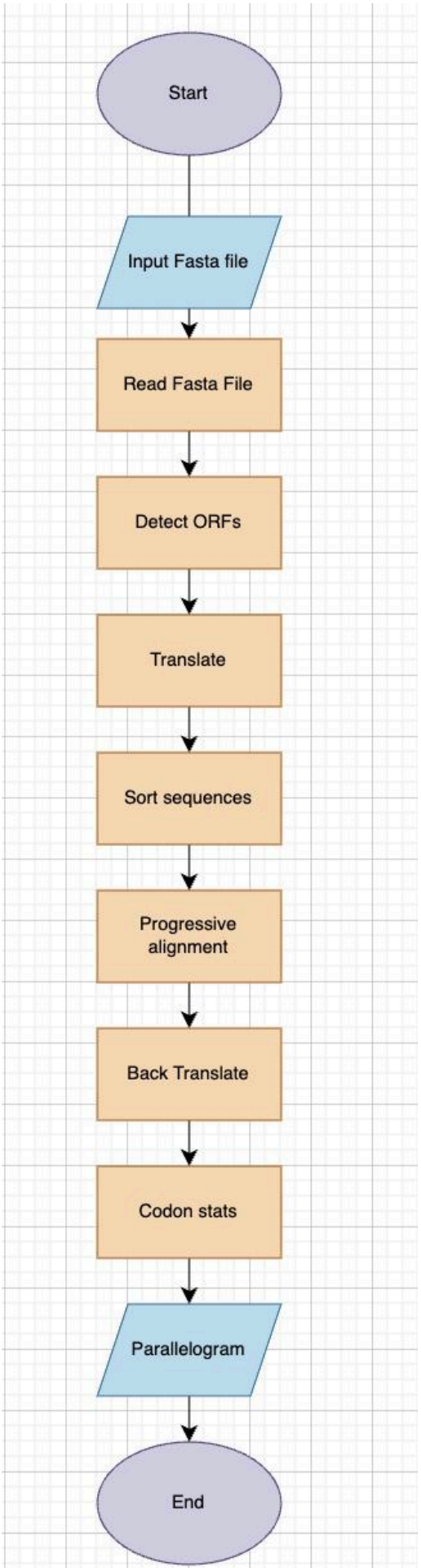
Function Title: main_pipeline

Description:
Runs the entire translation-based multiple sequence alignment workflow from start to finish.

Input 1: Input FASTA file
Input 2: Output directory
Output: Aligned FASTA files, CSV stats, plots
(Potential) Output: Error messages for invalid paths

Process:
Step 1: Read input FASTA.
Step 2: Detect ORFs.
Step 3: Translate ORFs to amino acids.

Step 4: Compute k-mer similarity matrix.
Step 5: Sort sequences.
Step 6: Perform progressive AA alignment.
Step 7: Back-translate amino acid alignment to nucleotide alignment.
Step 8: Compute codon position statistics.
Step 9: Generate and save plots.
Step 10: Save all outputs and terminate.


Start
Input Fasta file
Read Fasta File
Detect ORFs
Translate
Sort sequences
Progressive alignment
Back Translate
Codon stats
Parallelogram
End